

Zhenyu Lei

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Work Experience:

Independent Researcher | Cyber Security Association of China Project Jun 2025 – Present

Conduct independent research through a Student Innovation Research Project on processor microarchitectural behavior modeling and evaluation.

Synthesized recent findings in processor microarchitecture, extracted common behavioral patterns, and developed a prototype workflow for microarchitectural modeling and cross-platform benchmark evaluation.

Research Assistant | Southeast University, Nanjing, China Mar 2025 – Present

Conduct research on processor microarchitectural testing and robustness evaluation using feedback-driven black-box testing techniques under the supervision of Associate Professor Fei Tong.

Designed and implemented a semantics-guided testing framework to improve instruction-space coverage and evaluate processor robustness under noisy execution environments.

Research Assistant | National University of Singapore, Singapore Dec 2025 – Apr 2026

Conducted research on contract-based formal verification of processor timing behavior under the supervision of Assistant Professor Flavien Solt.

Formulated a hierarchical timing-contract model and developed a decoupled formal verification framework for separate processor-side and platform-side checks, enabling reuse of proof results.

Independent Researcher | Institute of Computing Technology, Chinese Academy of Sciences, Beijing, China
Sep 2022 – Jun 2023

Conducted an independent processor design project on an in-order, five-stage pipelined RISC-V processor.

Implemented core processor modules, optimized five-stage pipeline control, and built a differential testing workflow against a reference emulator to validate cycle-level correctness on complex workloads.

Education:

Southeast University, Nanjing, China

Master of Engineering in Cyberspace Security | Sep 2023 – Jun 2026 expected

Relevant Coursework: Pattern Recognition, Machine Learning, Matrix Theory for Engineering, Advanced Topics in Cyberspace Security

Southeast University, Nanjing, China

Bachelor of Engineering in Cyber Science (Honors Program for Gifted Youth) | Sep 2019 – Jun 2023

Relevant Coursework: Computer Architecture, Operating Systems, Pattern Recognition, Databases, Distributed Systems, Algorithms, Compilation Method, Probability and Statistics

Publications and Patents:

[1] Zhenyu Lei, Fei Tong, Xiaoyu Cheng and Guang Cheng. “Feedback-Driven Black-Box Testing for Processor Microarchitecture”, under review.

[2] Zhenyu Lei, Fei Tong, Xiaoyu Cheng and Flavien Solt. “Timing-Contract-Based Decoupled Formal Verification for Processor Timing Behavior”, under review.

Awards, Interests, Skills, and Community Involvement:

Awards:

First Place | National Undergraduate Extracurricular Science & Technology Competition (2021)

Second Prize (Cadence Track) | China Postgraduate Integrated Circuit Innovation Competition (2025)

Interests:

Computer architecture, memory systems, high-performance computing, compilers, automated hardware and software testing.

Professional Skills:

Programming: C, C++, Rust, Python, Assembly, Verilog, SystemVerilog, Scala, Tcl

Tools: gem5, Chipyard, Cadence Genus, Cadence Innovus, Cadence Joules, Cadence JasperGold

Languages: English (Proficient), Mandarin (Native)